11 October 2026

The Editors
Nature Machine Intelligence

Re: Reusability report: Tabular foundation models lead in cancer survival prediction

Dear Editors,

We submit our manuscript for consideration as a Reusability Report in Nature Machine Intelligence. It examines the reuse of the published TabPFN implementation from Hollmann and colleagues, “Accurate predictions on small data with a tabular foundation model” (Nature, 2025; doi:10.1038/s41586-024-08328-6).

TabPFN promises accurate predictions on small tables without tuning, which is exactly the situation of most clinical cohorts. We asked whether that promise holds when the code is taken out of its benchmark and applied to survival prediction from cancer multiomics, and what a high score in that setting actually means.

On six datasets from the original benchmark, TabPFN again had the highest mean ROC AUC among default-setting models (0.877 against 0.873 for CatBoost); this reproduces the direction of the original claim under a simpler protocol, not its effect size. We then evaluated four TabPFN generations, Google's TabFM, AutoGluon and five classical learners on 3,238 patients from five TCGA and CPTAC cancer cohorts, using 400 frozen test sets shared by every model. Within single cancers, a foundation model had the best mean ROC AUC in 7 of 13 tasks and TabPFN v2.5 had the best mean rank, although survival was hard for every method and no two nonlinear models differed significantly.

The second half of the report explains why this matters beyond one model. Pooling the five cancers raised ROC AUC to 0.721 for 3-year survival, but tree ensembles matched that score, and controls that knew only the cancer type or the pattern of zero-valued features came close to it. Scored within each cancer, pooled training gave no consistent benefit, and on extreme survival it lowered the C-index of foundation models by 0.022 more than that of tree ensembles. A pooled pan-cancer score can therefore look like a gain in prediction while mostly reflecting which cancer a patient has.

We believe the work fits the Reusability Report format because it shows both where the published code remains strong and where its scores need a different reading. TabPFN ran without tuning and was consistently among the best models; our recommendation is that pooled results be reported alongside within-cancer results and simple shortcut controls. We also report openly that an earlier version of one estimate pooled predictions across test sets, and we show the corrected estimate.

All code is available at https://github.com/HawkFranklin-Research/tab-r1-experiments, together with the 400 frozen test sets, per-patient predictions for every model and the source data for every figure and table, so each reported number can be regenerated without refitting a model. An archived copy of these data is deposited on figshare under the reserved DOI 10.6084/m9.figshare.34332477, which becomes active on publication; until then it can be viewed through a private link for the editor: https://figshare.com/s/ea59fbc60b0bc75591e1. We have also prepared an anonymised version of the manuscript in case double-blind review is preferred; in that version the repository link is withheld and shared with you only, and reviewers may request access through the editorial office.

This manuscript is original, has not been published, and is not under consideration at another journal. All authors have approved its submission, and the authors declare no competing interests.

Thank you for considering this report.

Yours sincerely,

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Authors are listed in manuscript order. This page is provided for the editor, because the anonymised version of the manuscript omits names, affiliations and contributions.

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2	Satya Prakash Mohanty	HawkFranklin Research, India	Conceived the study; performed the reproduction experiments; interpreted the results; wrote the manuscript.
3	Bikram Bhakat	Department of Zoology, The University of Burdwan, Purba Bardhaman, West Bengal, India	Reviewed and edited the manuscript; contributed to data visualization.
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Language-model assistance was used for code development, editing and reproducibility checks under human supervision; such tools are not authors and bear no responsibility for the conclusions. All authors approved the submitted version.